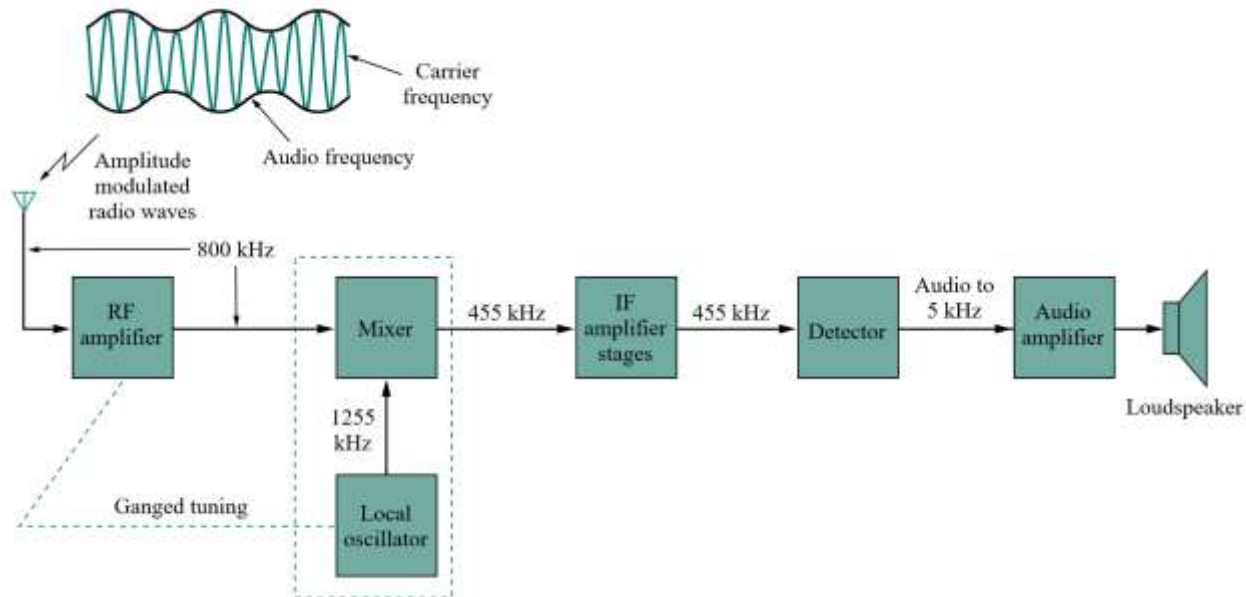
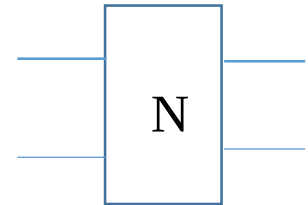
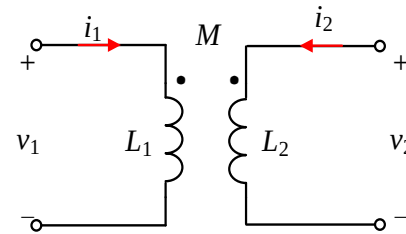
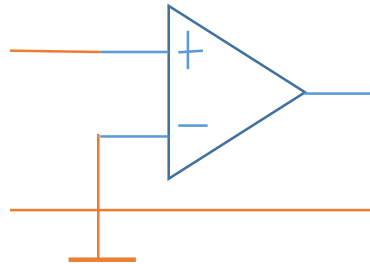
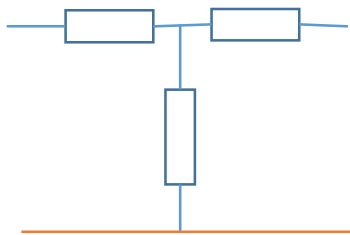
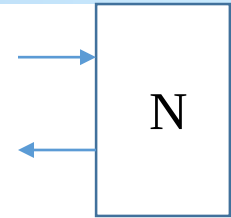


Advanced AC Circuit

Two-Port Networks



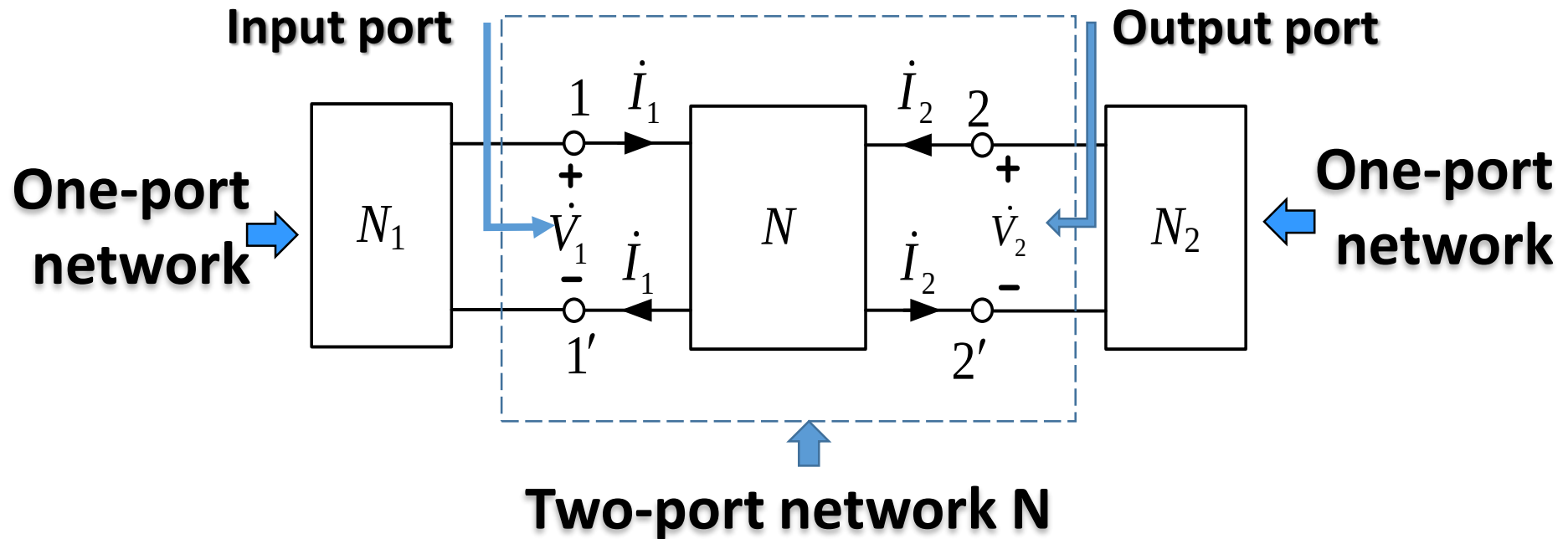
Outline

- **Parameters for two-port network analysis**
- **Two-port network analysis**
- **The connection of two-port networks**
- **Analysis for complicated two-port networks**

Parameters for two-port network analysis

- Introduction

A two-port network is an electrical network with two separate ports for input and output.

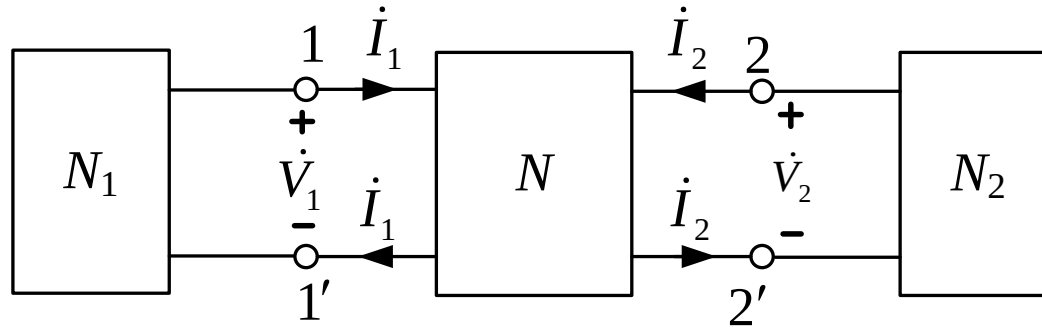


Assuming :

- (1) no independent source;
- (2) excited by DC or steady-state AC

Parameters for two-port network analysis

Types of Transfer Function



Driving point function

$$Z_{11} = \left. \frac{\dot{V}_1}{\dot{I}_1} \right|_{\text{condition at port}}$$

$$Y_{11} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\text{condition at port}}$$

Transfer function

$$H_I = \left. \frac{\dot{I}_2}{\dot{I}_1} \right|_{\text{condition at port}}$$

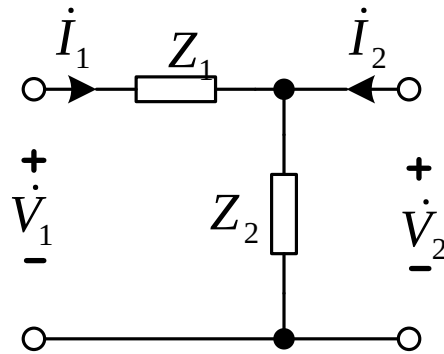
$$Z_{21} = \left. \frac{\dot{V}_2}{\dot{I}_1} \right|_{\text{condition at port}}$$

$$Y_{21} = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\text{condition at port}}$$

$$H_V = \left. \frac{\dot{V}_2}{\dot{V}_1} \right|_{\text{condition at port}}$$

Parameters for two-port network analysis

Example $\dot{V}_1$ is the input. Find



Solution:

$$Y_{11}|_{\dot{I}_2=0} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{I}_2=0} = \frac{1}{Z_1 + Z_2}$$

$$Y_{11}|_{\dot{V}_2=0} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{V}_2=0} = \frac{1}{Z_1}$$

$$Y_{11}|_{\dot{I}_2=0} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{I}_2=0}$$

$$Y_{11}|_{\dot{V}_2=0} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{V}_2=0}$$

$$H_v|_{\dot{I}_2=0} = \left. \frac{\dot{V}_2}{\dot{V}_1} \right|_{\dot{I}_2=0}$$

$$Y_{21}|_{\dot{V}_2=0} = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\dot{V}_2=0}$$

$$H_v|_{\dot{I}_2=0} = \left. \frac{\dot{V}_2}{\dot{V}_1} \right|_{\dot{I}_2=0} = \frac{Z_2}{Z_1 + Z_2}$$

$$Y_{21}|_{\dot{V}_2=0} = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\dot{V}_2=0} = -\frac{1}{Z_1}$$

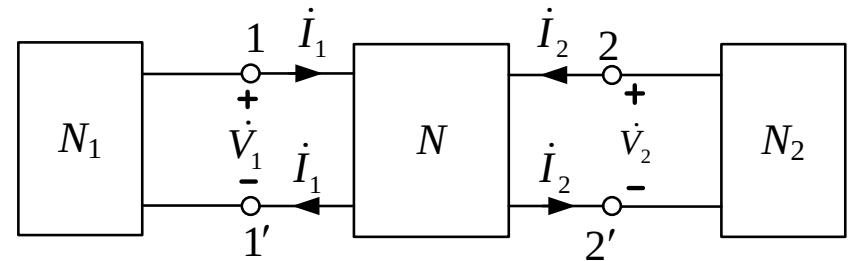
Note:

1. Transfer functions are different when input and output are in different positions.
2. Transfer function varies with condition at port.

Parameters for two-port network analysis

Parameters:

- The various terms that relate voltages and currents at ports are called parameters.



- Only one transfer function cannot characterize the two-port network
- Two out of four transfer equations are needed to characterize the four variables at ports — $C_4^2 = 6$ set parameters are available.

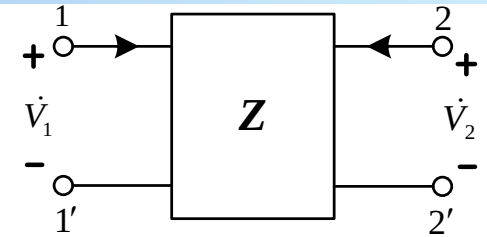
Impedance Parameters —Z parameters

Z parameter equations

$$\dot{V}_1 = z_{11}\dot{I}_1 + z_{12}\dot{I}_2$$

$$\dot{V}_2 = z_{21}\dot{I}_1 + z_{22}\dot{I}_2$$

$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \mathbf{Z} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} \Rightarrow \mathbf{Z} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$$



Z matrix

Methods to determine Z (and also Y, H, G, T, T') parameters:

(1) by analysis

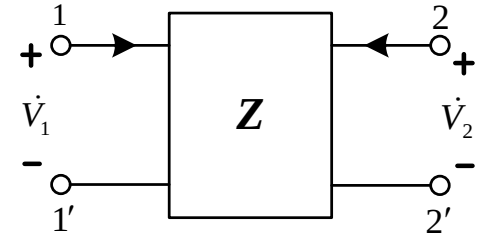
(2) by measurement

$$z_{11} = \left. \frac{\dot{V}_1}{\dot{I}_1} \right|_{\dot{I}_2=0}, \quad z_{12} = \left. \frac{\dot{V}_1}{\dot{I}_2} \right|_{\dot{I}_1=0}, \quad z_{21} = \left. \frac{\dot{V}_2}{\dot{I}_1} \right|_{\dot{I}_2=0}, \quad z_{22} = \left. \frac{\dot{V}_2}{\dot{I}_2} \right|_{\dot{I}_1=0}$$

Impedance Parameters —Z parameters

$$Z_{11} = \left. \frac{\dot{V}_1}{\dot{I}_1} \right|_{\dot{I}_2=0}, \quad Z_{12} = \left. \frac{\dot{V}_1}{\dot{I}_2} \right|_{\dot{I}_1=0}, \quad Z_{21} = \left. \frac{\dot{V}_2}{\dot{I}_1} \right|_{\dot{I}_2=0}, \quad Z_{22} = \left. \frac{\dot{V}_2}{\dot{I}_2} \right|_{\dot{I}_1=0}$$

Z — open-circuit impedance parameters



Z_{11} = open-circuit input impedance

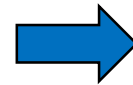
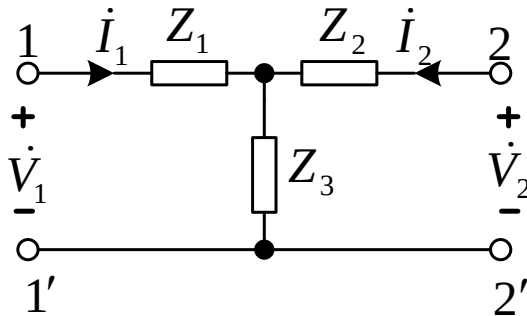
Z_{12} = open-circuit transfer impedance from port 1 to port 2

Z_{21} = open-circuit transfer impedance from port 2 to port 1

Z_{22} = open-circuit output impedance

Impedance Parameters

Example Find the T type two-ports **Z** parameter matrix.



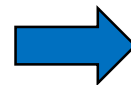
$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

$$\mathbf{Z} = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_2 + Z_3 \end{bmatrix}$$



Solution 1 : By mesh analysis

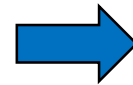
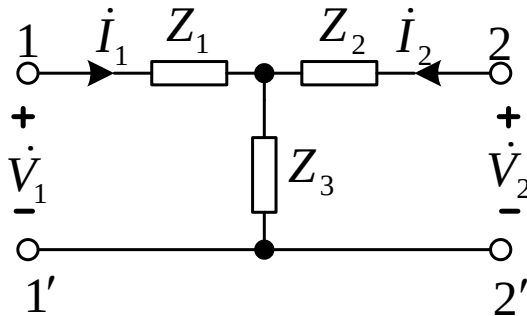
$$\begin{cases} \dot{V}_1 = (Z_1 + Z_3)\dot{I}_1 + Z_3\dot{I}_2 \\ \dot{V}_2 = Z_3\dot{I}_1 + (Z_2 + Z_3)\dot{I}_2 \end{cases}$$



$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_2 + Z_3 \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

Impedance Parameters

Example Find the T type two-ports **Z** parameter matrix.

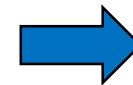


$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

Solution 2 : By measurement

$$z_{11} = \left. \frac{\dot{V}_1}{\dot{I}_1} \right|_{\dot{I}_2=0} = Z_1 + Z_3 \quad z_{12} = \left. \frac{\dot{V}_1}{\dot{I}_2} \right|_{\dot{I}_1=0} = Z_3$$

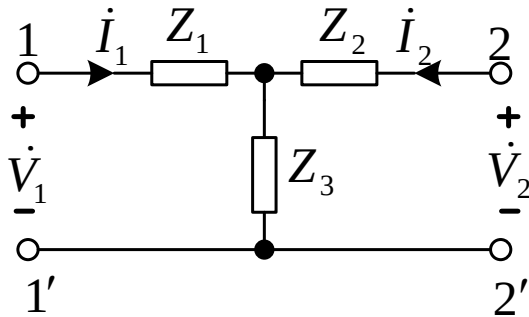
$$z_{21} = \left. \frac{\dot{V}_2}{\dot{I}_1} \right|_{\dot{I}_2=0} = Z_3 \quad z_{22} = \left. \frac{\dot{V}_2}{\dot{I}_2} \right|_{\dot{I}_1=0} = Z_2 + Z_3$$



$$\mathbf{Z} = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_2 + Z_3 \end{bmatrix}$$



Impedance Parameters



$$\mathbf{Z} = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_2 + Z_3 \end{bmatrix}$$



Discussion : (1) When $Z_1 = Z_2$,

$$\mathbf{Z} = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_1 + Z_3 \end{bmatrix}$$

It is a symmetrical network

$$Z_{11} = Z_{22}$$

(2) For a two-port network that is linear and has no dependent sources, the transfer impedances are equal

$$Z_{12} = Z_{21}$$

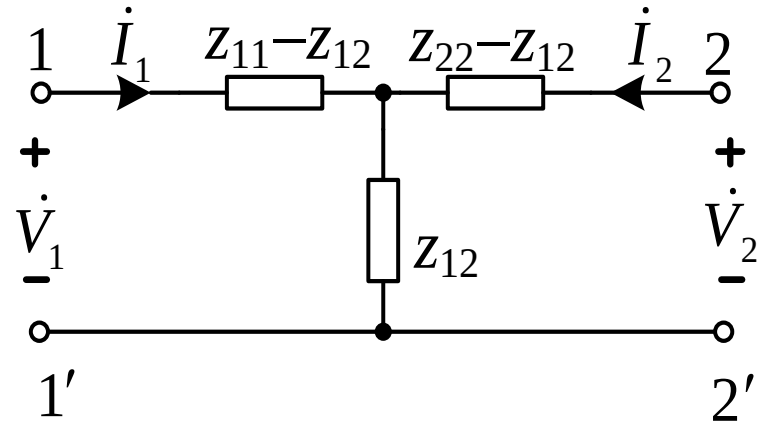
and the two-port is said to be **reciprocal**.

Impedance Parameters

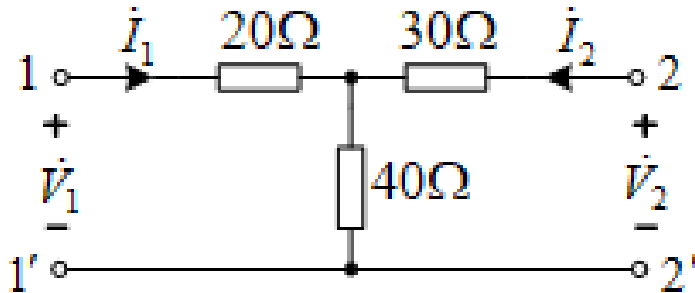
Discussion (3)

$$\dot{V}_1 = z_{11}\dot{I}_1 + z_{12}\dot{I}_2$$

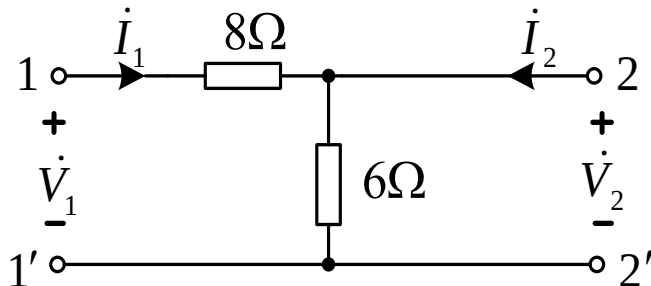
$$\dot{V}_2 = z_{21}\dot{I}_1 + z_{22}\dot{I}_2$$



EX.



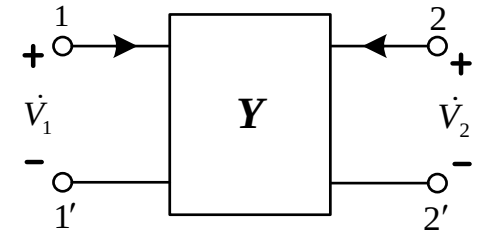
$$\mathbf{Z} = \begin{bmatrix} 60 & 40 \\ 40 & 70 \end{bmatrix}$$



$$\mathbf{Z} = \begin{bmatrix} 14 & 6 \\ 6 & 6 \end{bmatrix}$$

Admittance Parameters — Y parameters

Y parameter equations

$$\begin{aligned} \dot{I}_1 &= y_{11}\dot{V}_1 + y_{12}\dot{V}_2 \\ \dot{I}_2 &= y_{21}\dot{V}_1 + y_{22}\dot{V}_2 \end{aligned}$$


$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \mathbf{Y} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} \Rightarrow \text{Y matrix } \mathbf{Y} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$$

$$y_{11} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{V}_2=0}, \quad y_{12} = \left. \frac{\dot{I}_1}{\dot{V}_2} \right|_{\dot{V}_1=0}, \quad y_{21} = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\dot{V}_2=0}, \quad y_{22} = \left. \frac{\dot{I}_2}{\dot{V}_2} \right|_{\dot{V}_1=0} \quad \mathbf{Y} = \mathbf{Z}^{-1}$$

Y — short-circuit admittance parameters

y_{11} = short-circuit input admittance

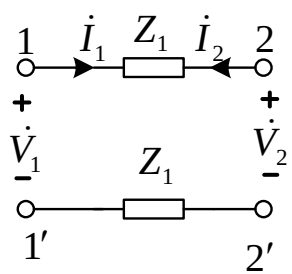
y_{12} = short-circuit transfer admittance from port 2 to port 1

y_{21} = short-circuit transfer admittance from port 1 to port 2

y_{22} = short-circuit output admittance

Admittance Parameters

Example Find Y matrix



$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2Z_1} & -\frac{1}{2Z_1} \\ -\frac{1}{2Z_1} & \frac{1}{2Z_1} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix}$$

Solution:

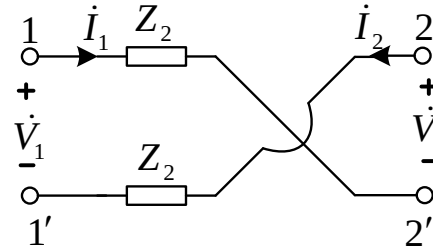
By measurement

$$Y = \begin{bmatrix} \frac{1}{2Z_1} & -\frac{1}{2Z_1} \\ -\frac{1}{2Z_1} & \frac{1}{2Z_1} \end{bmatrix}$$

$$y_{11} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{V}_2=0} = \frac{1}{2Z_1} \quad y_{12} = \left. \frac{\dot{I}_1}{\dot{V}_2} \right|_{\dot{V}_1=0} = -\frac{1}{2Z_1}$$

$$y_{21} = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\dot{V}_2=0} = -\frac{1}{2Z_1} \quad y_{22} = \left. \frac{\dot{I}_2}{\dot{V}_2} \right|_{\dot{V}_1=0} = \frac{1}{2Z_1}$$

$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix}$$



$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2Z_2} & \frac{1}{2Z_2} \\ \frac{1}{2Z_2} & \frac{1}{2Z_2} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix}$$

$$Y = \begin{bmatrix} \frac{1}{2Z_2} & \frac{1}{2Z_2} \\ \frac{1}{2Z_2} & \frac{1}{2Z_2} \end{bmatrix}$$

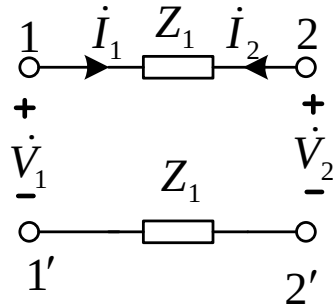
$$y_{11} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{V}_2=0} = \frac{1}{2Z_2}$$

$$y_{21} = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\dot{V}_2=0} = \frac{1}{2Z_2}$$

$$y_{12} = \left. \frac{\dot{I}_1}{\dot{V}_2} \right|_{\dot{V}_1=0} = \frac{1}{2Z_2}$$

$$y_{22} = \left. \frac{\dot{I}_2}{\dot{V}_2} \right|_{\dot{V}_1=0} = \frac{1}{2Z_2}$$

Admittance Parameters



$$\mathbf{Y} = \begin{bmatrix} \frac{1}{2Z_1} & -\frac{1}{2Z_1} \\ -\frac{1}{2Z_1} & \frac{1}{2Z_1} \end{bmatrix}$$

Discussion : (1) For a **Symmetrical network**

$$y_{11} = y_{22}$$

(2) For a two-port network that is linear and has no dependent sources, the transfer admittances are equal

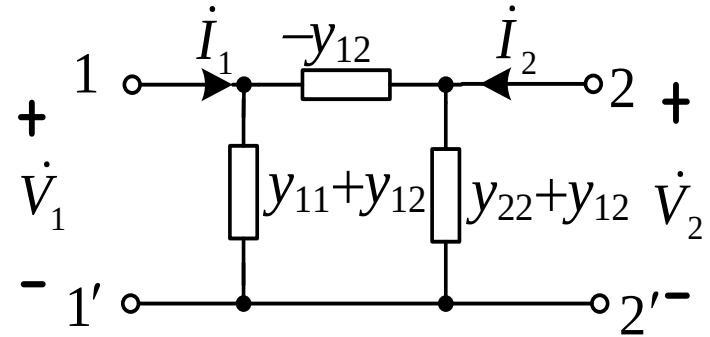
$$y_{12} = y_{21}$$

— **reciprocal**

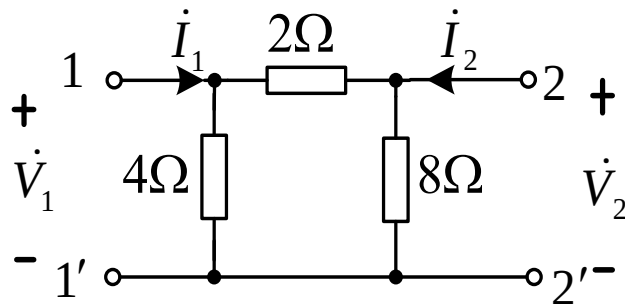
Admittance Parameters

Discussion : (3)

$$\begin{aligned} \dot{I}_1 &= y_{11}\dot{V}_1 + y_{12}\dot{V}_2 \\ \dot{I}_2 &= y_{21}\dot{V}_1 + y_{22}\dot{V}_2 \end{aligned} \quad \Rightarrow$$



EX.



$$\mathbf{Y} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{8} \end{bmatrix}$$

$$y_{12} = -\frac{1}{2} \text{ S}$$

$$y_{11} + y_{12} = \frac{1}{4} \text{ S} \quad \Rightarrow \quad y_{11} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \text{ S}$$

$$y_{22} + y_{12} = \frac{1}{8} \text{ S} \quad \Rightarrow \quad y_{22} = \frac{1}{8} + \frac{1}{2} = \frac{5}{8} \text{ S}$$

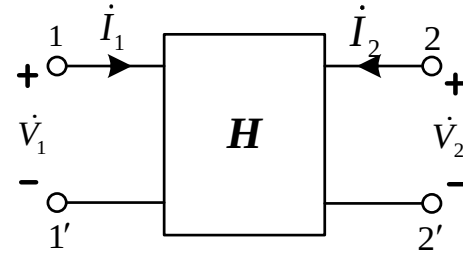
Hybrid Parameters

H parameter equation

$$\dot{V}_1 = h_{11}\dot{I}_1 + h_{12}\dot{V}_2$$

$$\dot{I}_2 = h_{21}\dot{I}_1 + h_{22}\dot{V}_2$$

$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} = \mathbf{H} \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix}$$



H matrix $\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$

$$h_{11} = \left. \frac{\dot{V}_1}{\dot{I}_1} \right|_{\dot{V}_2=0}, \quad h_{12} = \left. \frac{\dot{V}_1}{\dot{V}_2} \right|_{\dot{I}_1=0}, \quad h_{21} = \left. \frac{\dot{I}_2}{\dot{I}_1} \right|_{\dot{V}_2=0}, \quad h_{22} = \left. \frac{\dot{I}_2}{\dot{V}_2} \right|_{\dot{I}_1=0}$$

h_{11} = short-circuit input impedance

h_{12} = open-circuit reverse voltage gain

h_{21} = short-circuit forward current gain

h_{22} = open-circuit output admittance

For reciprocal networks,

$$h_{12} = -h_{21}.$$

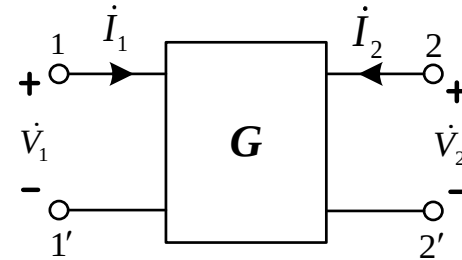
Reverse Hybrid Parameters

G parameter equation

$$\dot{I}_1 = g_{11}\dot{V}_1 + g_{12}\dot{I}_2$$

$$\dot{V}_2 = g_{21}\dot{V}_1 + g_{22}\dot{I}_2$$

$$\begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} = \mathbf{G} \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix}$$



G matrix $\mathbf{G} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$

$$g_{11} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{I}_2=0}, \quad g_{12} = \left. \frac{\dot{I}_1}{\dot{I}_2} \right|_{\dot{V}_1=0}, \quad g_{21} = \left. \frac{\dot{V}_2}{\dot{V}_1} \right|_{\dot{I}_2=0}, \quad g_{22} = \left. \frac{\dot{V}_2}{\dot{I}_2} \right|_{\dot{V}_1=0}$$

g_{11} = open-circuit input admittance

g_{12} = short-circuit reverse current gain

g_{21} = open-circuit forward voltage gain

g_{22} = short-circuit output impedance

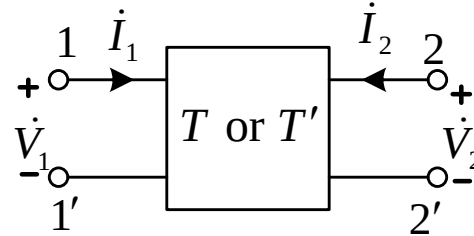
$$\mathbf{G} = \mathbf{H}^{-1}$$

Transmission Parameters

T parameter equation

$$\dot{V}_1 = A\dot{V}_2 + B(-\dot{I}_2)$$

$$\dot{I}_1 = C\dot{V}_2 + D(-\dot{I}_2)$$



➔
$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix} = \mathbf{T} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}$$

T matrix
$$\mathbf{T} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$A = \left. \frac{\dot{V}_1}{\dot{V}_2} \right|_{\dot{I}_2=0}, \quad B = \left. \frac{\dot{V}_1}{-\dot{I}_2} \right|_{\dot{V}_2=0}, \quad C = \left. \frac{\dot{I}_1}{\dot{V}_2} \right|_{\dot{I}_2=0}, \quad D = \left. \frac{\dot{I}_1}{-\dot{I}_2} \right|_{\dot{V}_2=0}$$

A = open-circuit voltage ratio

B = negative short-circuit transfer impedance

C = open-circuit transfer admittance

D = negative short-circuit current ratio

Inverse Transmission Parameters

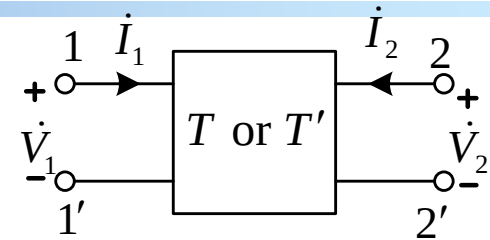
Inverse T parameter equation (T' or t)

$$\dot{V}_2 = A'\dot{V}_1 + B'(-\dot{I}_1)$$

$$\dot{I}_2 = C'\dot{V}_1 + D'(-\dot{I}_1)$$

$$\Rightarrow \begin{bmatrix} \dot{V}_2 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix} = \mathbf{T}' \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix}$$

$$A' = \left. \frac{\dot{V}_2}{\dot{V}_1} \right|_{\dot{I}_1=0}, \quad B' = \left. \frac{\dot{V}_2}{-\dot{I}_1} \right|_{\dot{V}_1=0}, \quad C' = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\dot{I}_1=0}, \quad D' = \left. \frac{\dot{I}_2}{-\dot{I}_1} \right|_{\dot{V}_1=0}$$



T' matrix

$$\mathbf{T}' = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix}$$

A' = Open-circuit voltage gain

B' = Negative short-circuit transfer impedance

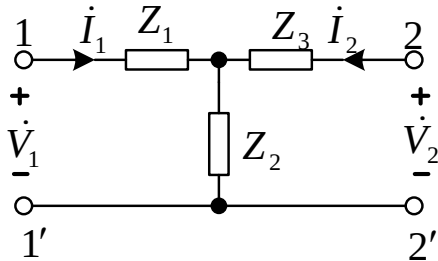
C' = Open-circuit transfer admittance

D' = Negative short-circuit current gain

$$\begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} = \frac{1}{AD - BC} \begin{bmatrix} D & B \\ C & A \end{bmatrix}$$

Transmission Parameters

Example Find T matrix



$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \frac{1}{Z_2} \begin{bmatrix} Z_1 + Z_2 & Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1 \\ 1 & Z_2 + Z_3 \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}$$



$$\mathbf{T} = \frac{1}{Z_2} \begin{bmatrix} Z_1 + Z_2 & Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1 \\ 1 & Z_2 + Z_3 \end{bmatrix}$$



$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix} = \mathbf{T} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}$$

Solution : By measurement

$$A = \left. \frac{\dot{V}_1}{\dot{V}_2} \right|_{\dot{I}_2=0} = \frac{Z_1 + Z_2}{Z_2} = 1 + \frac{Z_1}{Z_2}$$

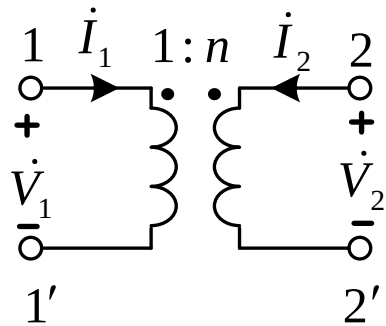
$$C = \left. \frac{\dot{I}_1}{\dot{V}_2} \right|_{\dot{I}_2=0} = \frac{1}{Z_2}$$

$$B = \left. \frac{\dot{V}_1}{(-\dot{I}_2)} \right|_{\dot{V}_2=0} = \frac{Z_3(-\dot{I}_2) + Z_1[(-\dot{I}_2) + \frac{Z_3(-\dot{I}_2)}{Z_2}]}{-\dot{I}_2} = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$D = \left. \frac{\dot{I}_1}{(-\dot{I}_2)} \right|_{\dot{V}_2=0} = \frac{Z_2 + Z_3}{Z_2} = 1 + \frac{Z_3}{Z_2}$$

Transmission Parameters

Example Find T and T' parameter equations



$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix} = \mathbf{T} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{V}_2 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix} = \mathbf{T}' \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix}$$

Solution: By measurement

$$\begin{cases} \dot{V}_1 / \dot{V}_2 = 1/n \\ \dot{I}_1 / \dot{I}_2 = -n \end{cases} \begin{matrix} \nearrow \\ \searrow \end{matrix} \begin{aligned} &\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} \frac{1}{n} & 0 \\ 0 & n \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix} = \mathbf{T} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix} \\ &\begin{bmatrix} \dot{V}_2 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & \frac{1}{n} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix} = \mathbf{T}' \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix} \end{aligned}$$

Recap of Parameters

$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \mathbf{Z} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

symmetrical

$$z_{11} = z_{22}$$

reciprocal

$$z_{12} = z_{21}$$

$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \mathbf{Y} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix}$$

$$\mathbf{Y} = \mathbf{Z}^{-1}$$

$$y_{11} = y_{22}$$

$$y_{12} = y_{21}$$

$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} = \mathbf{H} \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix}$$

$$h_{12} = -h_{21}$$

$$\mathbf{G} = \mathbf{H}^{-1}$$

$$\begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} = \mathbf{G} \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix} = \mathbf{T} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}$$

$$\begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} = \frac{1}{AD - BC} \begin{bmatrix} D & B \\ C & A \end{bmatrix}$$

$$AD - BC = 1$$

$$\begin{bmatrix} \dot{V}_2 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix} = \mathbf{T}' \begin{bmatrix} \dot{V}_1 \\ -\dot{I}_1 \end{bmatrix}$$

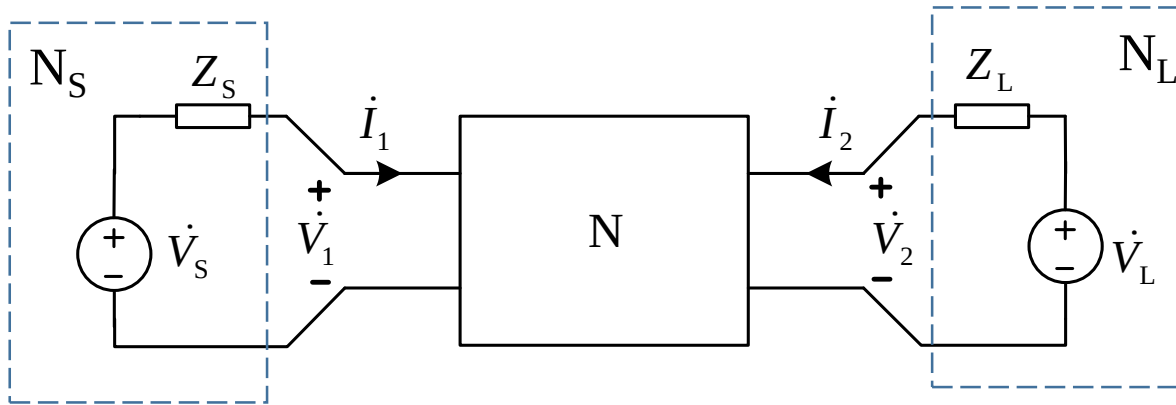
$$\mathbf{T}' \neq \mathbf{T}^{-1}$$

$$A'D' - B'C' = 1$$

-
- **Parameters for two-port network analysis**
 - **Two-port network analysis**
 - **The connections of two-port networks**
 - **Analysis for complicated two-port network**

Two-port network analysis

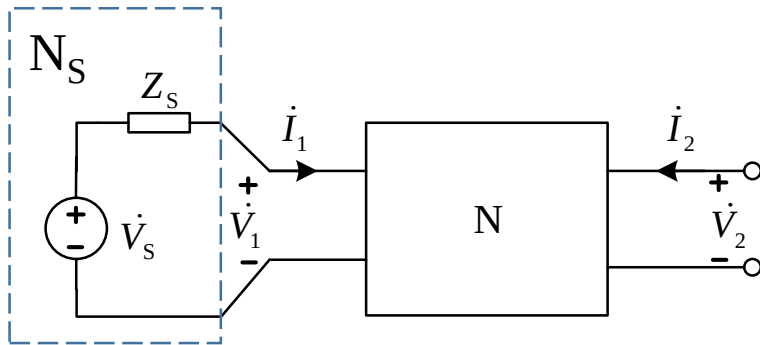
- Analysis for N when N_S and N_L are given,



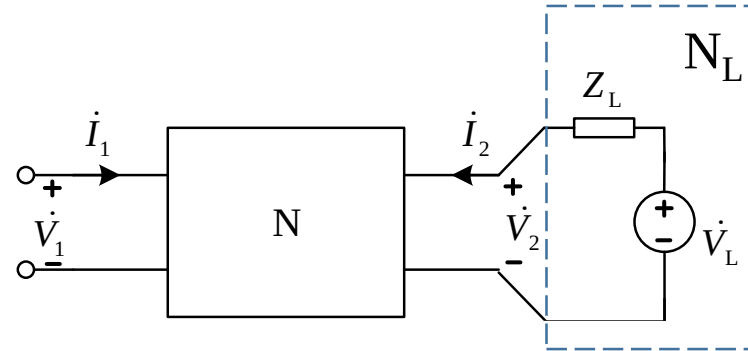
$$\begin{cases} \text{Parameter Equations} \\ \dot{V}_1 = \dot{V}_S - \dot{I}_1 Z_S \\ \dot{V}_2 = \dot{V}_L - \dot{I}_2 Z_L \end{cases}$$

Two-port network analysis

- Analysis for N when N_S or N_L are given



$$\begin{cases} \text{Parameter Equations} \\ \dot{V}_1 = \dot{V}_S - \dot{I}_1 Z_S \end{cases}$$



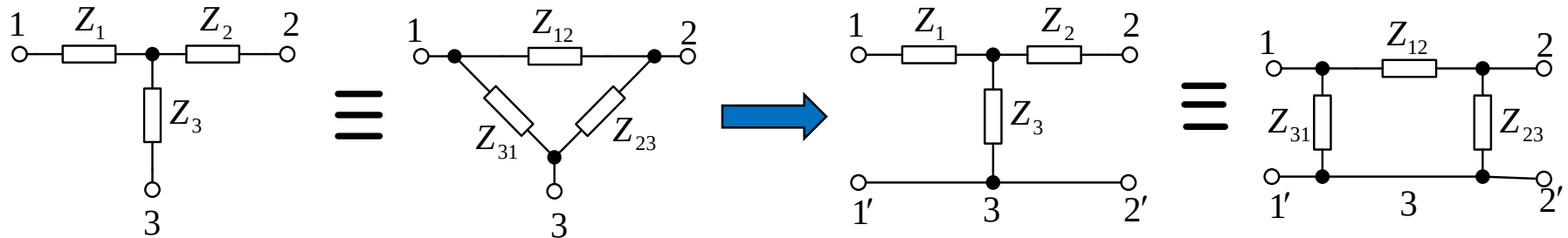
$$\begin{cases} \text{Parameter Equations} \\ \dot{V}_2 = \dot{V}_L - \dot{I}_2 Z_L \end{cases}$$

—Analysis for

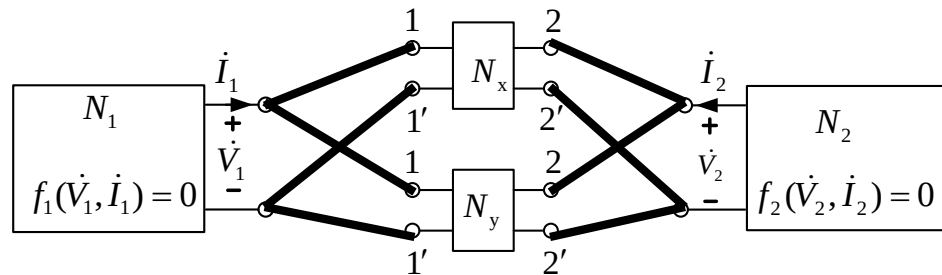
- equivalent input/output impedance,
- voltage/current gain,
- transfer impedance/admittance,
- power

Two-port network analysis

Example Find the relationships for equivalence



Solution

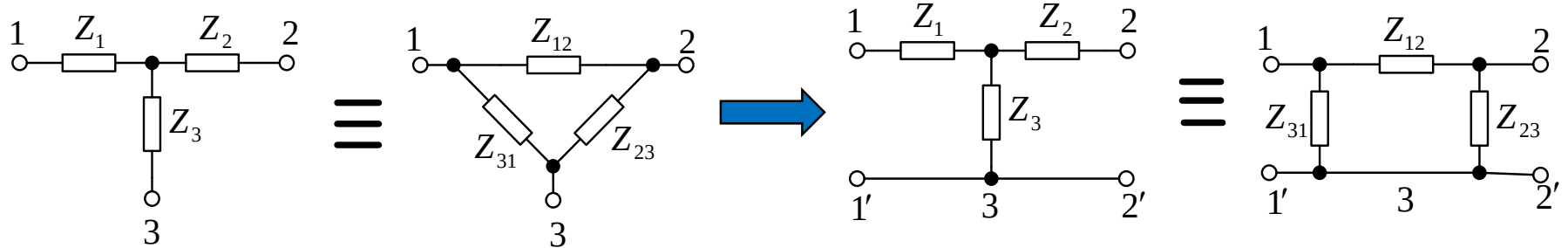


$$\left\{ \begin{array}{l} \text{Para. Equ. for } N_x \\ f_1(\dot{V}_1, \dot{I}_1) = 0 \\ f_2(\dot{V}_2, \dot{I}_2) = 0 \end{array} \right. \longleftrightarrow \left\{ \begin{array}{l} \text{Para. Equ. for } N_y \\ f_1(\dot{V}_1, \dot{I}_1) = 0 \\ f_2(\dot{V}_2, \dot{I}_2) = 0 \end{array} \right.$$

Equivalence: for any set of parameters

Two-port network analysis

Example Find the relationships for equivalence



Solution

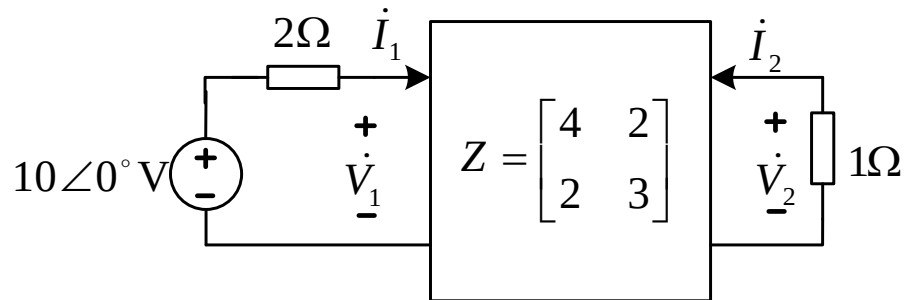
$$Z_T = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_2 + Z_3 \end{bmatrix} \longleftrightarrow Z_{\Pi} = \begin{bmatrix} \frac{Z_{31}(Z_{12} + Z_{23})}{Z_{12} + Z_{23} + Z_{31}} & \frac{Z_{23}Z_{31}}{Z_{12} + Z_{23} + Z_{31}} \\ \frac{Z_{23}Z_{31}}{Z_{12} + Z_{23} + Z_{31}} & \frac{Z_{23}(Z_{12} + Z_{31})}{Z_{12} + Z_{23} + Z_{31}} \end{bmatrix}$$

$$\begin{cases} Z_1 = \frac{Z_{31}Z_{12}}{Z_{12} + Z_{23} + Z_{31}} \\ Z_2 = \frac{Z_{12}Z_{23}}{Z_{12} + Z_{23} + Z_{31}} \\ Z_3 = \frac{Z_{23}Z_{31}}{Z_{12} + Z_{23} + Z_{31}} \end{cases}$$

$$\begin{cases} Z_{12} = \frac{Z_1Z_2 + Z_2Z_3 + Z_3Z_1}{Z_3} \\ Z_{23} = \frac{Z_1Z_2 + Z_2Z_3 + Z_3Z_1}{Z_1} \\ Z_{31} = \frac{Z_1Z_2 + Z_2Z_3 + Z_3Z_1}{Z_2} \end{cases}$$

Two-port network analysis

Example Find the variables at two ports.



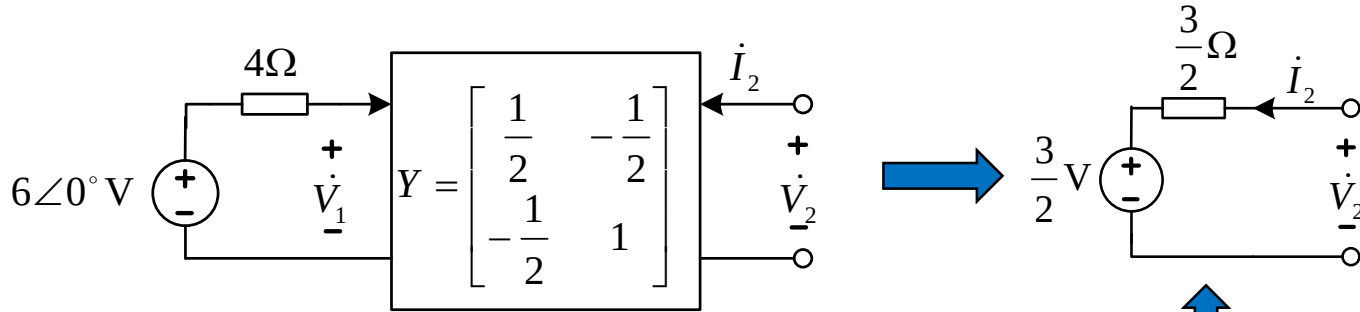
Solution

$$\begin{aligned} \dot{V}_1 &= 10 - 2\dot{I}_1 \\ \begin{cases} \dot{V}_1 &= 4\dot{I}_1 + 2\dot{I}_2 \\ \dot{V}_2 &= 2\dot{I}_1 + 3\dot{I}_2 \end{cases} \\ \dot{V}_2 &= -\dot{I}_2 \end{aligned}$$

$$\begin{cases} \dot{V}_1 = 6\text{ V}, \dot{I}_1 = 2\text{ A} \\ \dot{V}_2 = 1\text{ V}, \dot{I}_2 = -1\text{ A} \end{cases}$$

Two-port network analysis

Example Find the Thevinin equivalence at output port.

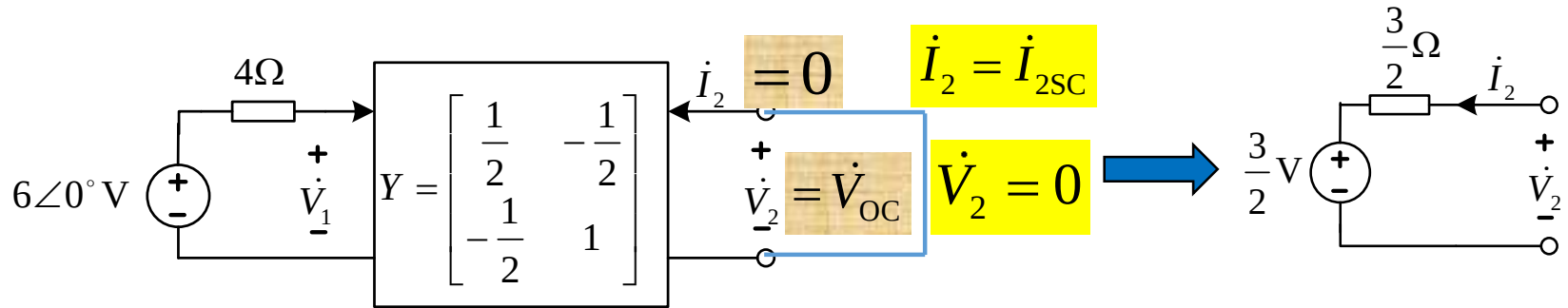


Solution 1

$$\begin{aligned} \dot{V}_1 &= 6 - 4\dot{I}_1 \\ \begin{cases} \dot{I}_1 &= \frac{1}{2}\dot{V}_1 - \frac{1}{2}\dot{V}_2 \\ \dot{I}_2 &= -\frac{1}{2}\dot{V}_1 + \dot{V}_2 \end{cases} \end{aligned} \quad \Rightarrow \quad \dot{V}_2 = \frac{3}{2} + \frac{3}{2}\dot{I}_2$$

Two-port network analysis

Example Find the Thevinin equivalence at output port.



Solution 2 Open circuit voltage

Equivalence impedance (open-short circuit method)

$$\begin{cases} \dot{I}_1 = \frac{1}{2}\dot{V}_1 - \frac{1}{2}\dot{V}_{2OC} \\ 0 = -\frac{1}{2}\dot{V}_1 + \dot{V}_{2OC} \\ \dot{V}_1 = 6 - 4\dot{I}_1 \end{cases} \Rightarrow \dot{V}_{2OC} = \frac{3}{2} \text{ V}$$

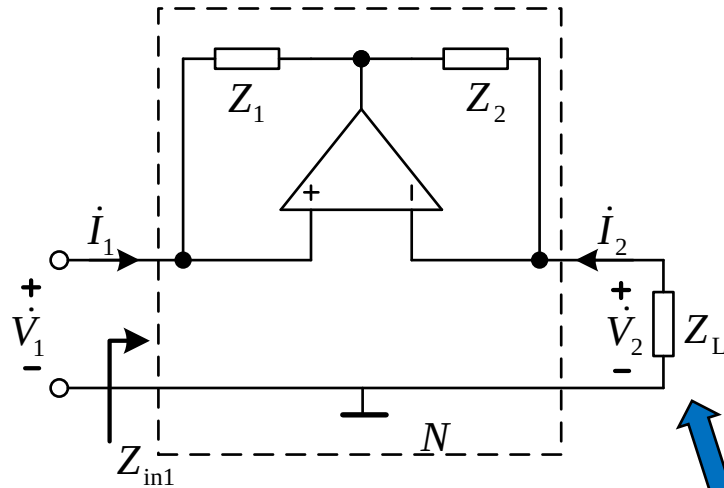
$$\dot{I}_{2SC} = -1 \text{ A}$$

$$\begin{cases} \dot{I}_1 = \frac{1}{2}\dot{V}_1 \\ \dot{I}_{2SC} = -\frac{1}{2}\dot{V}_1 \\ \dot{V}_1 = 6 - 4\dot{I}_1 \end{cases}$$

$$Z_o = -\frac{\dot{V}_{2OC}}{\dot{I}_{2SC}} = \frac{3}{2} \Omega$$

Two-port network analysis

Example Find the input impedance.



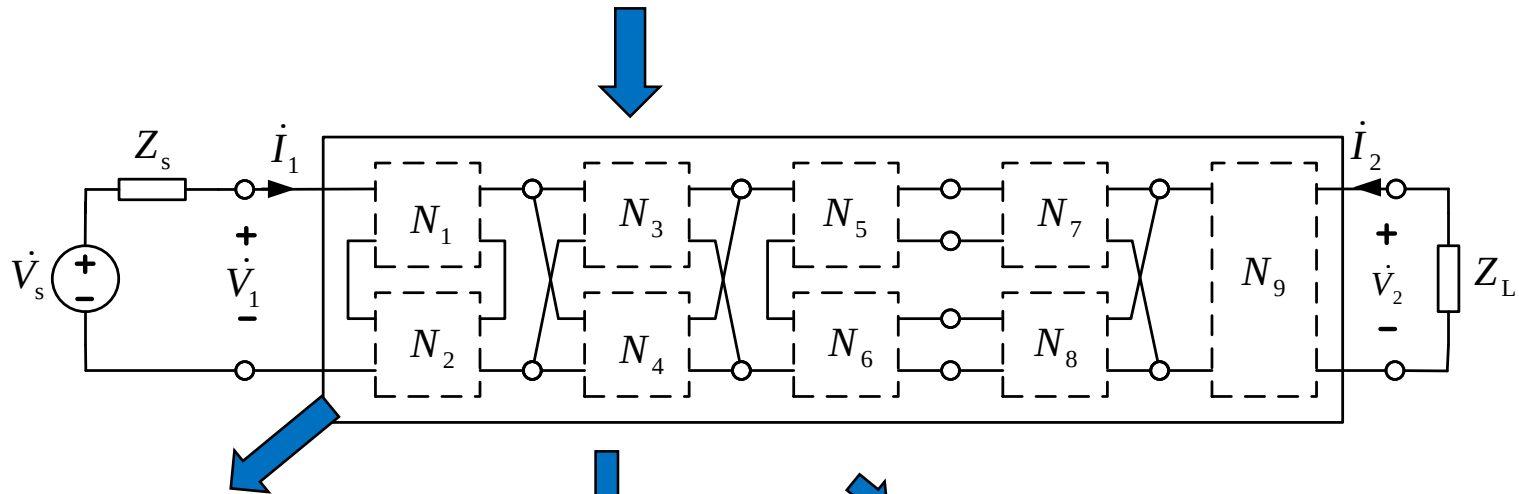
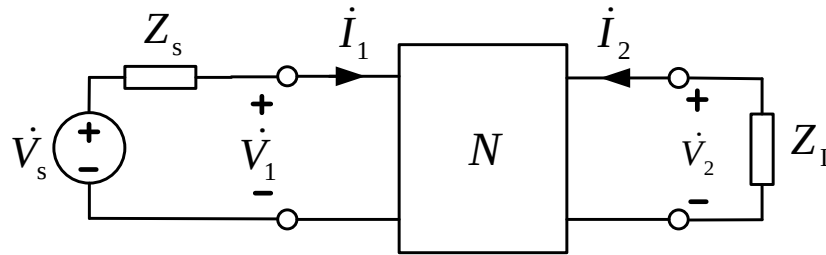
Solution

$$Z_{in1} = \frac{\dot{V}_1}{\dot{I}_1} = -\frac{Z_1}{Z_2} \frac{V_2}{(-I_2)} = -\frac{Z_1}{Z_2} Z_L$$

$$\begin{aligned} \dot{V}_1 &= \dot{V}_2 \\ \dot{I}_1 Z_1 &= \dot{I}_2 Z_2 \end{aligned} \quad \Rightarrow \quad \begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -\frac{Z_2}{Z_1} \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}$$

-
- **Parameters for two-port network analysis**
 - **Two-port network analysis**
 - **Interconnection of two-port networks**
 - **Analysis for complicated two-port network**

Interconnection of two-port networks



**N1 and N2
in series(+)**

**N3 and N4 in
parallel(//)**

**N5 and N7 in cascade
N6 and N8 in cascade(*)**

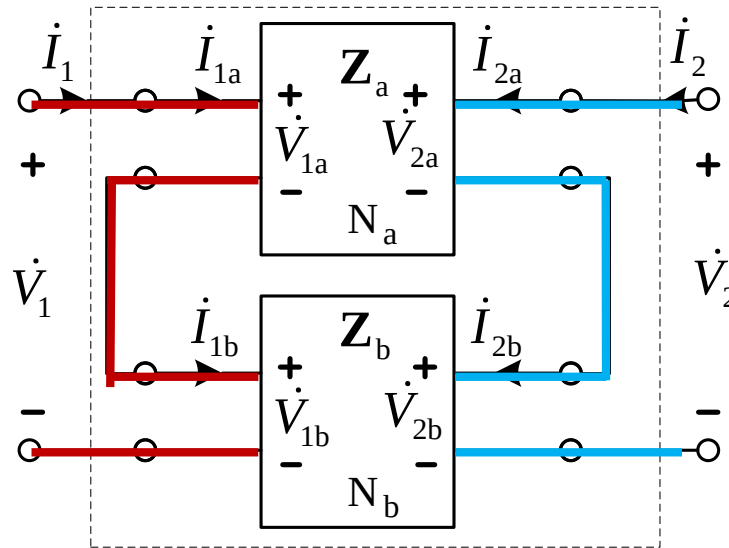
$(N_1 + N_2) * (N_3 // N_4)$

$(N_5 * N_7) + // (N_6 * N_8)$

Interconnection — In Series

KCL, KVL

$$\begin{cases} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} \dot{I}_{1a} \\ \dot{I}_{2a} \end{bmatrix} = \begin{bmatrix} \dot{I}_{1b} \\ \dot{I}_{2b} \end{bmatrix} \\ \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} \dot{V}_{1a} \\ \dot{V}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{V}_{1b} \\ \dot{V}_{2b} \end{bmatrix} \end{cases}$$



Para. Equ.

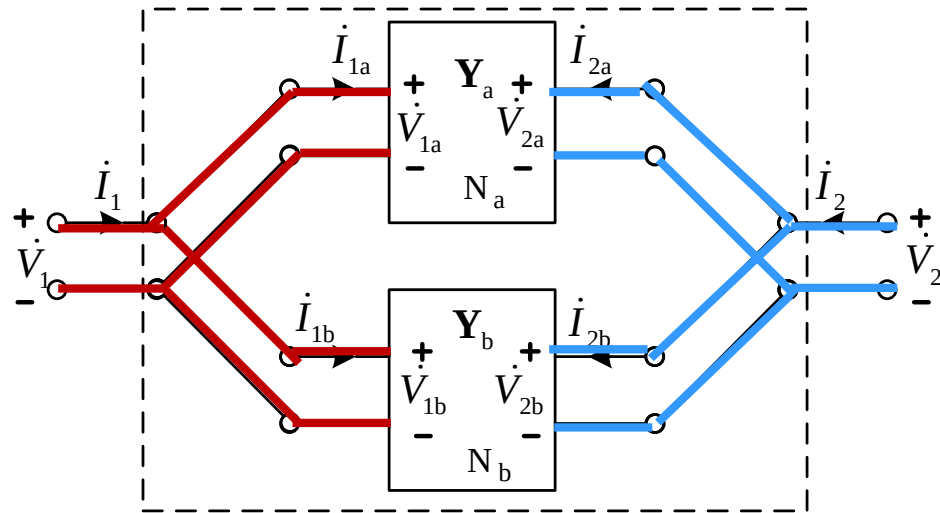
$$\begin{aligned} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} &= \begin{bmatrix} \dot{V}_{1a} \\ \dot{V}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{V}_{1b} \\ \dot{V}_{2b} \end{bmatrix} = \begin{bmatrix} Z_{11a} & Z_{12a} \\ Z_{21a} & Z_{22a} \end{bmatrix} \begin{bmatrix} \dot{I}_{1a} \\ \dot{I}_{2a} \end{bmatrix} + \begin{bmatrix} Z_{11b} & Z_{12b} \\ Z_{21b} & Z_{22b} \end{bmatrix} \begin{bmatrix} \dot{I}_{1b} \\ \dot{I}_{2b} \end{bmatrix} \\ &= (\mathbf{Z}_a + \mathbf{Z}_b) \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \mathbf{Z} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} \end{aligned}$$

Conclusion: For two ports in series $\mathbf{Z} = \mathbf{Z}_a + \mathbf{Z}_b$

Interconnection — In Parallel

KCL, KVL

$$\begin{cases} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} \dot{V}_{1a} \\ \dot{V}_{2a} \end{bmatrix} = \begin{bmatrix} \dot{V}_{1b} \\ \dot{V}_{2b} \end{bmatrix} \\ \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} \dot{I}_{1a} \\ \dot{I}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{I}_{1b} \\ \dot{I}_{2b} \end{bmatrix} \end{cases}$$



Para. Equ.

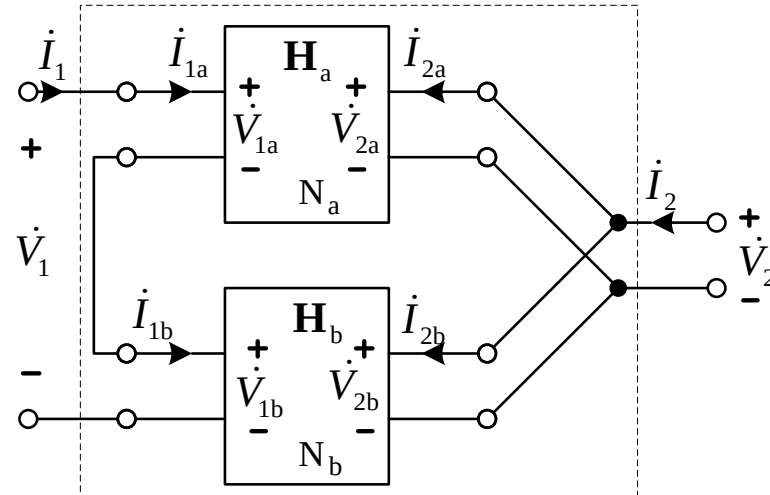
$$\begin{aligned} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} &= \begin{bmatrix} \dot{I}_{1a} \\ \dot{I}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{I}_{1b} \\ \dot{I}_{2b} \end{bmatrix} = \begin{bmatrix} y_{11a} & y_{12a} \\ y_{21a} & y_{22a} \end{bmatrix} \begin{bmatrix} \dot{V}_{1a} \\ \dot{V}_{2a} \end{bmatrix} + \begin{bmatrix} y_{11b} & y_{12b} \\ y_{21b} & y_{22b} \end{bmatrix} \begin{bmatrix} \dot{V}_{1b} \\ \dot{V}_{2b} \end{bmatrix} \\ &= (Y_a + Y_b) \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = Y \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} \end{aligned}$$

Conclusion: For two ports in parallel $Y = Y_a + Y_b$

Interconnection — In Series-Parallel

KCL, KVL

$$\begin{cases} \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} \dot{I}_{1a} \\ \dot{V}_{2a} \end{bmatrix} = \begin{bmatrix} \dot{I}_{1b} \\ \dot{V}_{2b} \end{bmatrix} \\ \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{V}_{1b} \\ \dot{I}_{2b} \end{bmatrix} \end{cases}$$



Para. Equ.

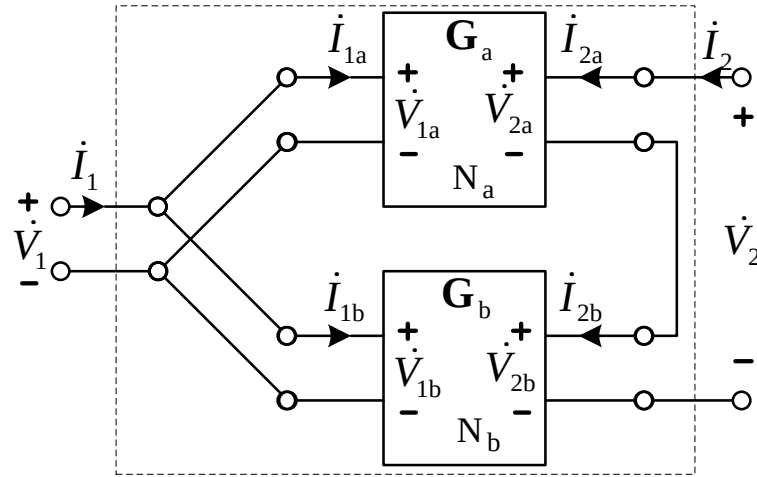
$$\begin{aligned} \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} &= \begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{V}_{1b} \\ \dot{I}_{2b} \end{bmatrix} = \begin{bmatrix} h_{11a} & h_{12a} \\ h_{21a} & h_{22a} \end{bmatrix} \begin{bmatrix} \dot{I}_{1a} \\ \dot{V}_{2a} \end{bmatrix} + \begin{bmatrix} h_{11b} & h_{12b} \\ h_{21b} & h_{22b} \end{bmatrix} \begin{bmatrix} \dot{I}_{1b} \\ \dot{V}_{2b} \end{bmatrix} \\ &= (H_a + H_b) \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} = H \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} \end{aligned}$$

Conclusion: For two ports in series-parallel $\mathbf{H} = \mathbf{H}_a + \mathbf{H}_b$

Interconnection — In Parallel-Series

KCL, KVL

$$\begin{cases} \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{2a} \end{bmatrix} = \begin{bmatrix} \dot{V}_{1b} \\ \dot{I}_{2b} \end{bmatrix} \\ \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} \dot{I}_{1a} \\ \dot{V}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{I}_{1b} \\ \dot{V}_{2b} \end{bmatrix} \end{cases}$$



Para. Equ.

$$\begin{aligned} \begin{bmatrix} \dot{I}_1 \\ \dot{V}_2 \end{bmatrix} &= \begin{bmatrix} \dot{I}_{1a} \\ \dot{V}_{2a} \end{bmatrix} + \begin{bmatrix} \dot{I}_{1b} \\ \dot{V}_{2b} \end{bmatrix} = \begin{bmatrix} g_{11a} & g_{12a} \\ g_{21a} & g_{22a} \end{bmatrix} \begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{2a} \end{bmatrix} + \begin{bmatrix} g_{11b} & g_{12b} \\ g_{21b} & g_{22b} \end{bmatrix} \begin{bmatrix} \dot{V}_{1b} \\ \dot{I}_{2b} \end{bmatrix} \\ &= (G_a + G_b) \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} = G \begin{bmatrix} \dot{V}_1 \\ \dot{I}_2 \end{bmatrix} \end{aligned}$$

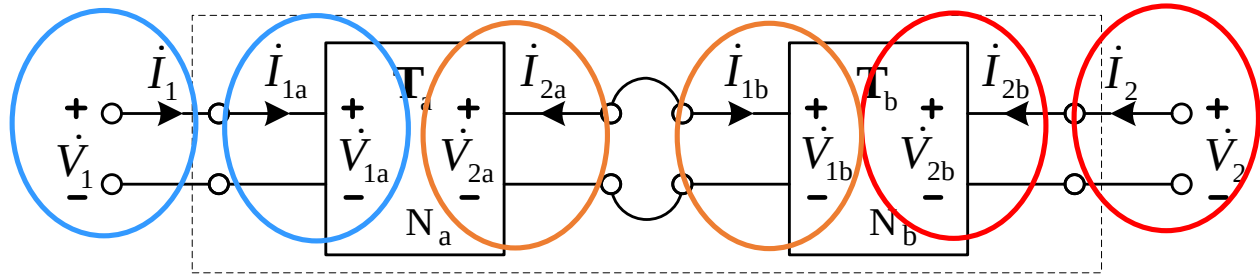
Conclusion: For two ports in parallel-series $G = G_a + G_b$

Interconnection — In Cascade

KCL, KVL

$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{1a} \end{bmatrix}$$

$$\begin{bmatrix} \dot{V}_{2a} \\ -\dot{I}_{2a} \end{bmatrix} = \begin{bmatrix} \dot{V}_{1b} \\ \dot{I}_{1b} \end{bmatrix}$$



$$\begin{bmatrix} \dot{V}_2 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} \dot{V}_{2b} \\ \dot{I}_{2b} \end{bmatrix}$$

$$\begin{bmatrix} \dot{V}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} \dot{V}_{2a} \\ -\dot{I}_{2a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} \dot{V}_{1b} \\ \dot{I}_{1b} \end{bmatrix}$$

$$= \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} \dot{V}_{2b} \\ -\dot{I}_{2b} \end{bmatrix}$$

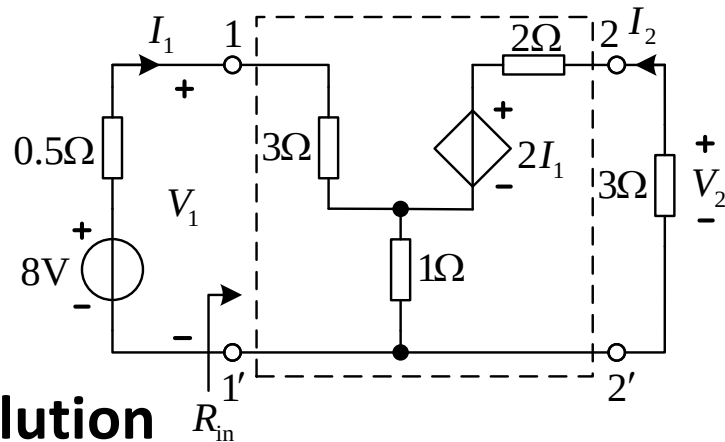
$$= (T_a T_b) \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix} = T \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}$$

Conclusion: For two ports in cascade $T = T_a T_b$

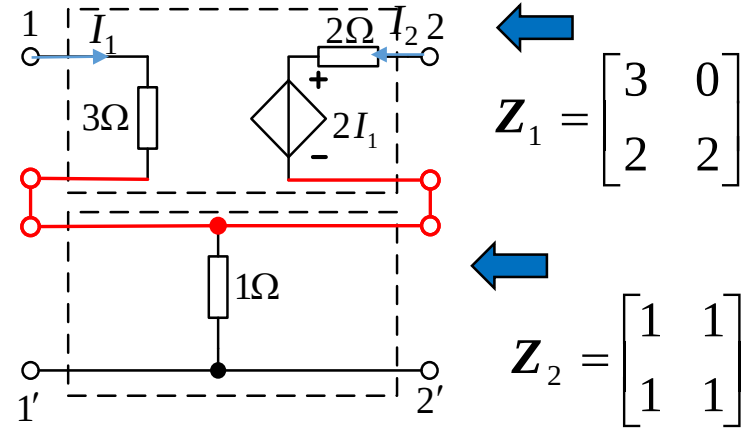
-
- **Parameters for two-port network analysis**
 - **Two-port network analysis**
 - **The connections of two-port networks**
 - **Analysis for complicated two-port network**

Analysis for complicated two-port network

Example Find R_{in} , I_1 , V_1 , I_2 and V_2 .



In series



$$\mathbf{Z}_1 = \begin{bmatrix} 3 & 0 \\ 2 & 2 \end{bmatrix}$$

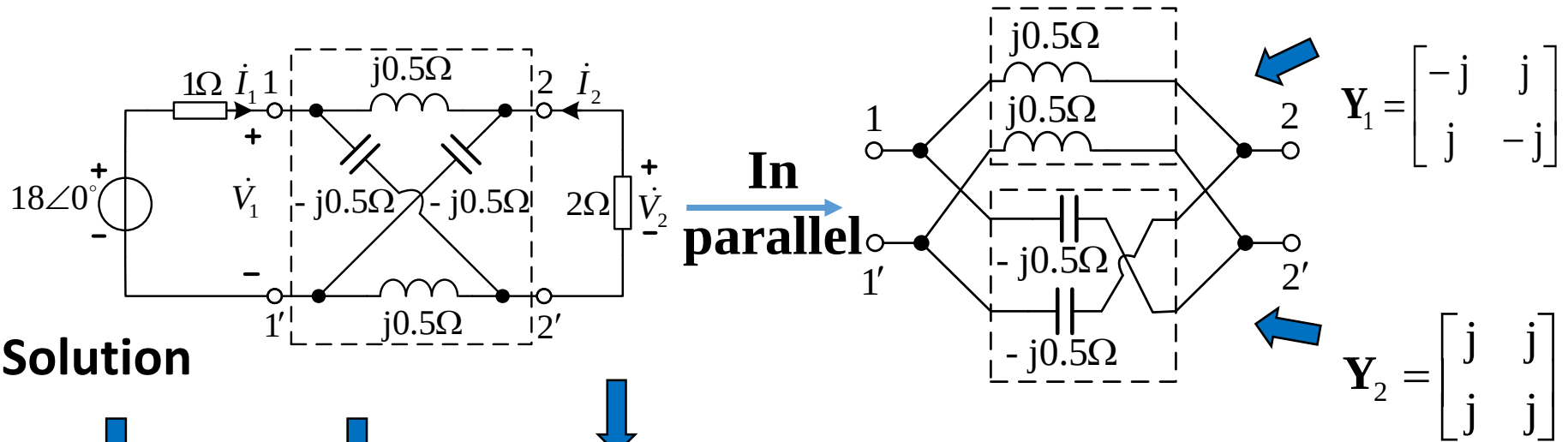
$$\mathbf{Z}_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Solution

$$\begin{aligned}
 & \downarrow \quad \downarrow \quad \downarrow \\
 V_1 &= 8 - 0.5I_1 & \mathbf{Z} &= \mathbf{Z}_1 + \mathbf{Z}_2 & V_2 &= -3I_2 \\
 & & &= \begin{bmatrix} 4 & 1 \\ 3 & 3 \end{bmatrix} & & \\
 & & \downarrow & & & \\
 \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} &= \begin{bmatrix} 4 & 1 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} & & & & \\
 & & & \searrow & & \\
 & & & R_{in} = \frac{V_1}{I_1} = 3.5\Omega & & \\
 & & & \downarrow & & \\
 \begin{cases} I_1 = \frac{8}{0.5 + 3.5} = 2\text{A} \\ V_1 = 3.5I_1 = 7\text{V} \end{cases} & \rightarrow & \begin{cases} I_2 = -1\text{A} \\ V_2 = 3\text{V} \end{cases}
 \end{aligned}$$

Analysis for complicated two-port network

Example Find $\dot{I}_1$, $\dot{V}_1$, $\dot{I}_2$ and $\dot{V}_2$



Solution

$$\dot{V}_1 = 18 - \dot{I}_1$$

$$\mathbf{Y} = \mathbf{Y}_1 + \mathbf{Y}_2 = \begin{bmatrix} 0 & j2 \\ j2 & 0 \end{bmatrix}$$

$$\dot{V}_2 = -2\dot{I}_2$$

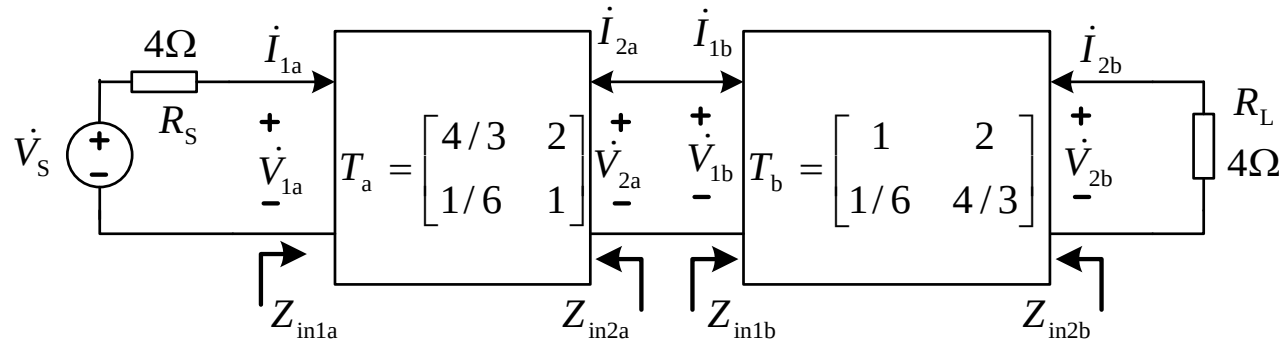
$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} 0 & j2 \\ j2 & 0 \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix}$$

$$\begin{cases} \dot{I}_1 = 16\text{A} \\ \dot{V}_1 = 2\text{V} \end{cases}$$

$$\begin{cases} \dot{I}_2 = j4\text{A} \\ \dot{V}_2 = -j8\text{V} \end{cases}$$

Analysis for complicated two-port network

Example Determine the total T parameter equations and prove all connections are match connections.



Solution

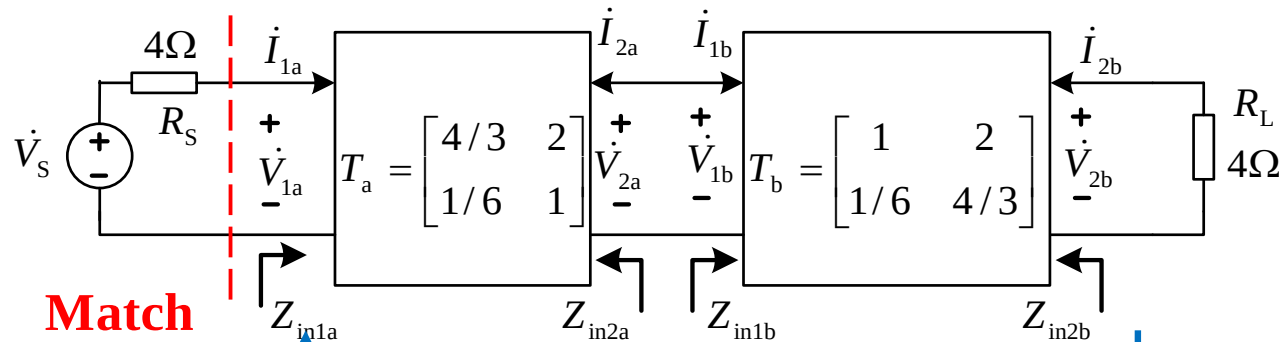
N_a and N_b are
connected in
cascade

$$\begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{1a} \end{bmatrix} = T_a T_b \begin{bmatrix} \dot{V}_{2b} \\ -\dot{I}_{2b} \end{bmatrix} = \begin{bmatrix} 5/3 & 16/3 \\ 1/3 & 5/3 \end{bmatrix} \begin{bmatrix} \dot{V}_{2b} \\ -\dot{I}_{2b} \end{bmatrix}$$

Analysis for complicated two-port network

Example ...prove all connections are match connections.

1. Prove Z_{1a} match R_s



$$\begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{1a} \end{bmatrix} = \begin{bmatrix} 5/3 & 16/3 \\ 1/3 & 5/3 \end{bmatrix} \begin{bmatrix} \dot{V}_{2b} \\ -\dot{I}_{2b} \end{bmatrix}$$

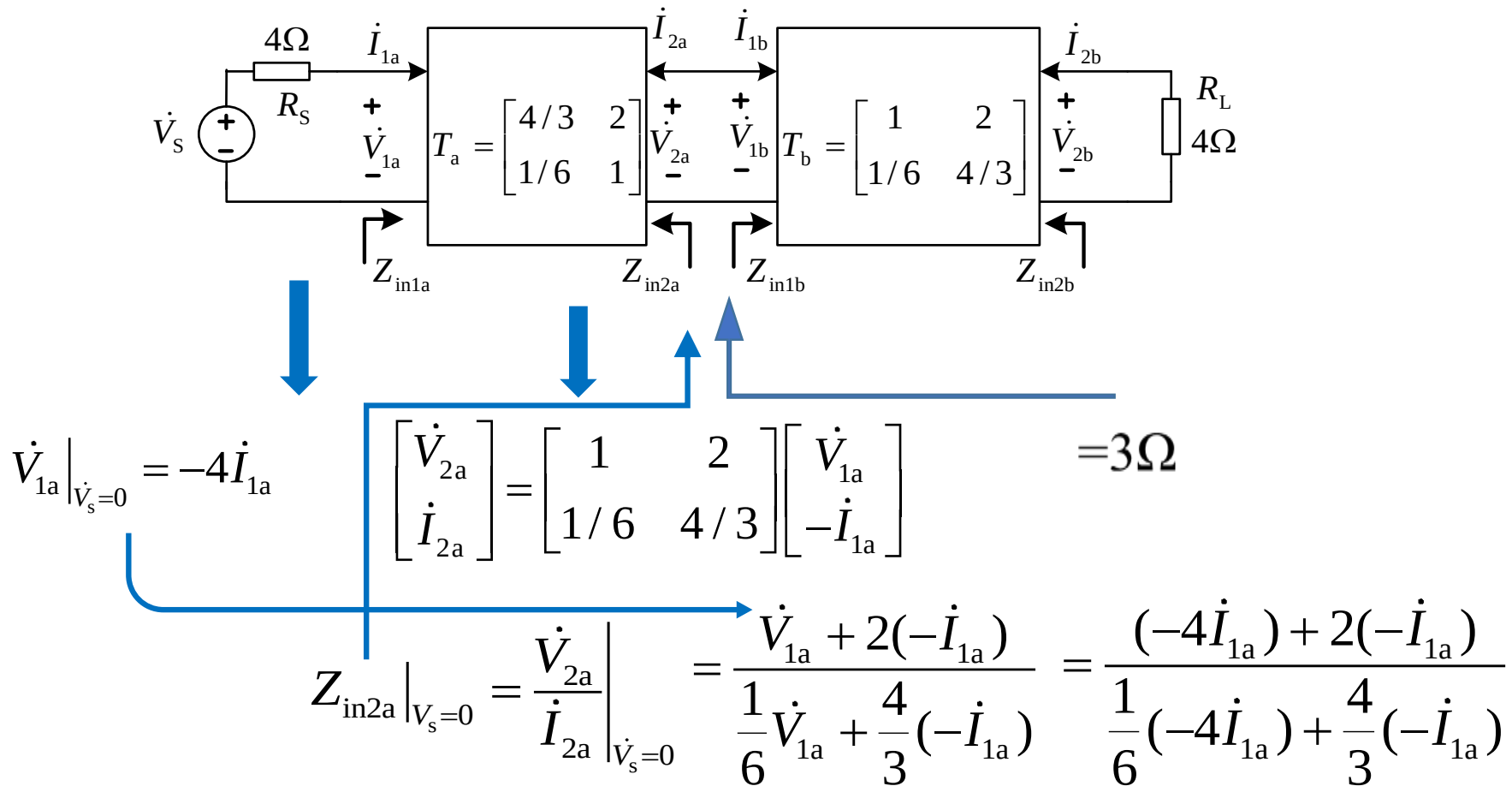
$$\dot{V}_{2b} = -4\dot{I}_{2b}$$

$$Z_{in1a} = \frac{\dot{V}_{1a}}{\dot{I}_{1a}} = \frac{\frac{5}{3}\dot{V}_{2b} + \frac{16}{3}(-\dot{I}_{2b})}{\frac{1}{3}\dot{V}_{2b} + \frac{5}{3}(-\dot{I}_{2b})} = \frac{\frac{5}{3}(-4\dot{I}_{2b}) + \frac{16}{3}(-\dot{I}_{2b})}{\frac{1}{3}(-4\dot{I}_{2b}) + \frac{5}{3}(-\dot{I}_{2b})} = 4\Omega = R_s$$

Analysis for complicated two-port network

Example ...prove all connections are match connections.

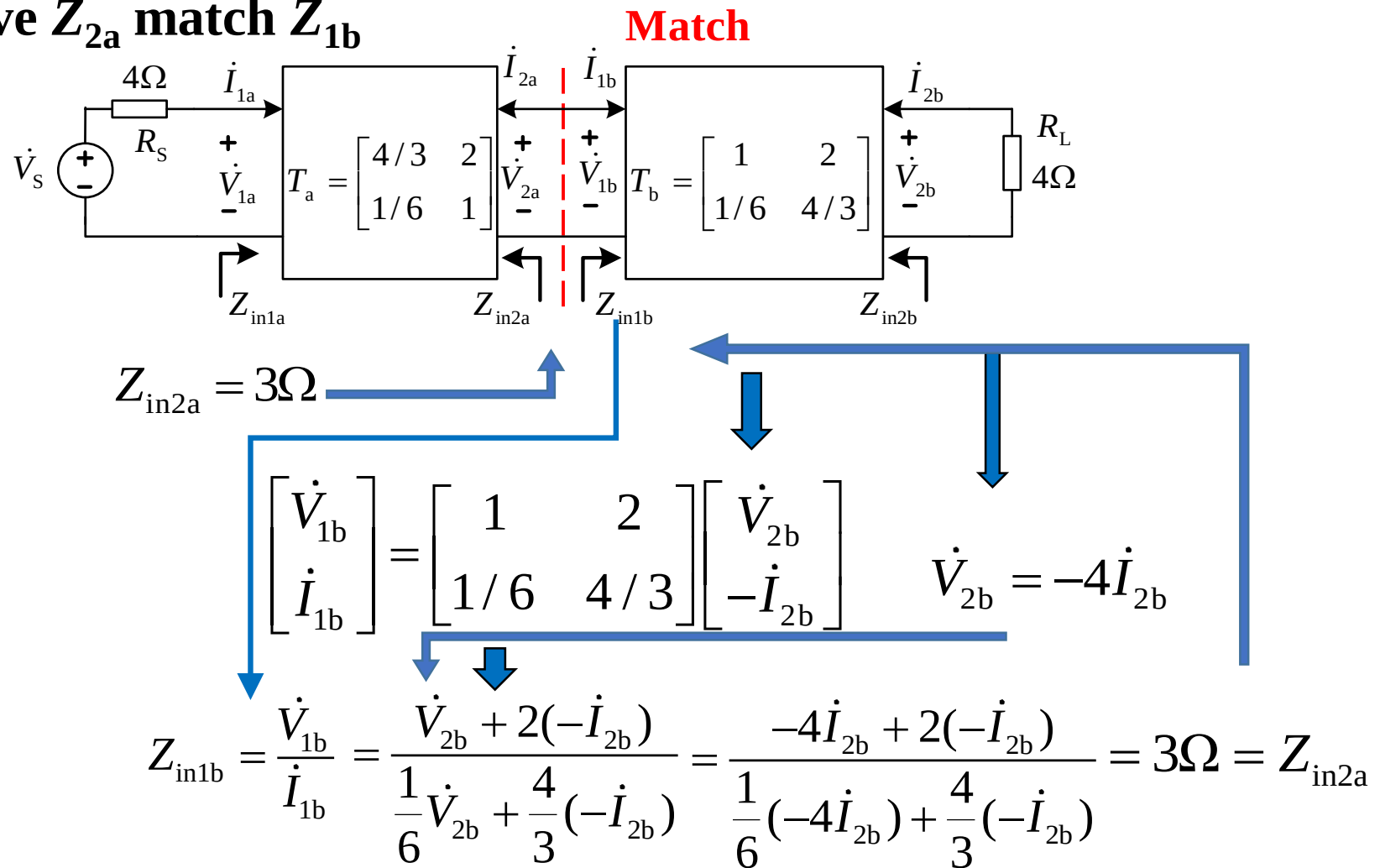
2. Prove Z_{2a} match Z_{1b}



Analysis for complicated two-port network

Example ...prove all connections are match connections.

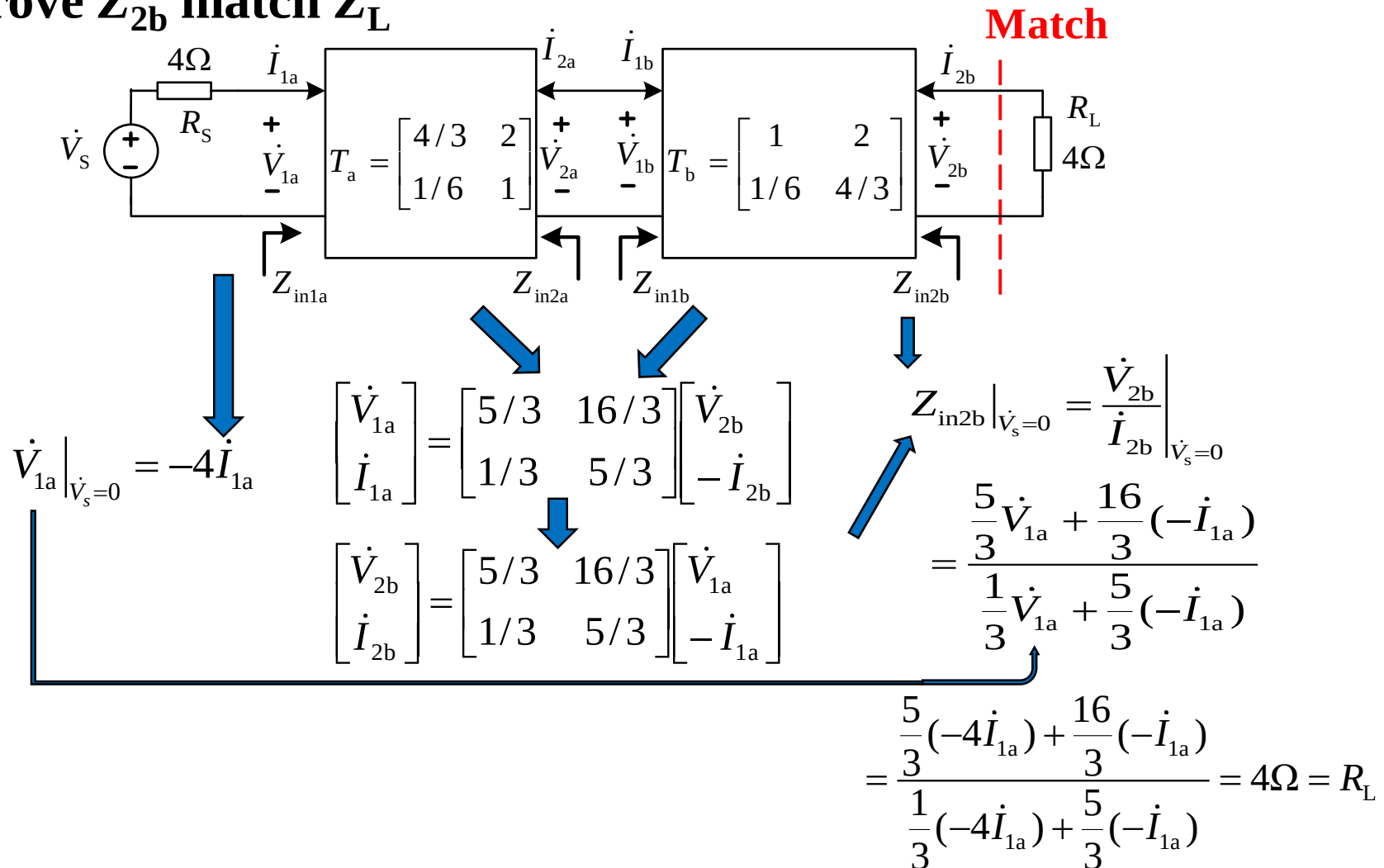
2. Prove Z_{2a} match Z_{1b}



Analysis for complicated two-port network

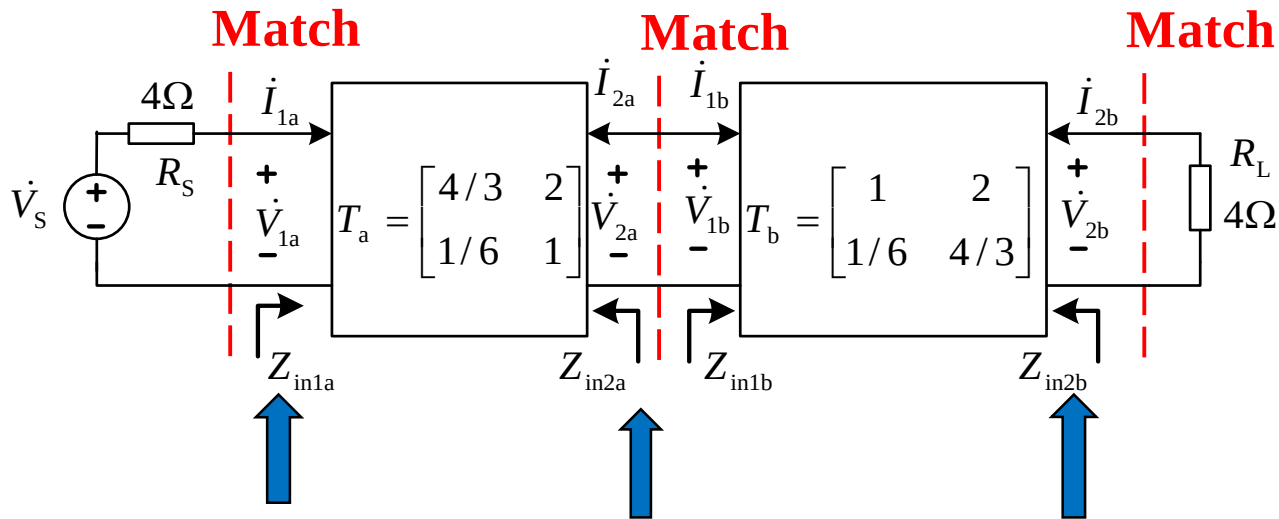
Example ...prove all connections are match connections.

3. Prove Z_{2b} match Z_L



Analysis for complicated two-port network

Example Determine the total T parameter equations and prove all connections are match connections.



$$Z_{in1a} = \frac{\dot{V}_{1a}}{\dot{I}_{1a}} = 4\Omega = R_s$$

$$Z_{in2a}|_{\dot{V}_s=0} = 3\Omega = Z_{in1b}$$

$$Z_{in2b}|_{\dot{V}_s=0} = 4\Omega = R_L$$

$$\begin{bmatrix} \dot{V}_{1a} \\ \dot{I}_{1a} \end{bmatrix} = \begin{bmatrix} 5/3 & 16/3 \\ 1/3 & 5/3 \end{bmatrix} \begin{bmatrix} \dot{V}_{2b} \\ -\dot{I}_{2b} \end{bmatrix}$$

Homework

- **4-1 (Z) , 4-2 (Y) , 4-3 (T) , 4-7 (Y)**
- **4-4 (T) , 4-8 (Z) , 4-9 (T) , 4-10 (equivalence)**
- **4-5 (parallel) , 4-12 (P) , 4-13 (S) , 4-14 (C)**